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# Privacy-Preserving and Quantum-Resistant Federated Learning Framework for Fog Healthcare Systems

**N. Kannan · K. Balasubramanian**

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**Abstract**
Quantum computing is on the horizon and will destroy existing cryptography standards, putting digital healthcare system security and patient privacy at danger. For very private and secure communication in fog healthcare settings, this article presents a new Quantum-Resistant Federated Deep Learning (QR-FDL) Framework. The QR-FDL architecture incorporates a lattice-based technique (like Kyber) into the model aggregation stage of Federated Learning (FL), which is a kind of Post-Quantum Cryptography (PQC). We provide a solution that uses Differential Privacy (DP) to prevent attacks based on gradients in data reconstruction and guarantees quantum-era security for weights of models with gradients sent between the central server and decentralized fog nodes. The practical use of QR-FDL is shown by its rapid convergence rate and excellent classification accuracy of 91.5% in an empirical assessment including a medical imaging job, such as tumor classification. Critically, we measure the cryptographic overhead and demonstrate that, even if PQC causes a regulated latency rise the total FL round time is still tolerable for real-time fog settings. In order to implement quantum-secure along with privacy-preserving Deep Learning in mission-critical healthcare communications, this study presents the first proven, end-to-end solution. In addition to showing that federated medical facilities can be securely encrypted from end to end, this architecture uses differential privacy during model training to protect patients' personal information. Through the research, we demonstrate its practical implementation in the near future for it is hard to balance the precision of modeled obstructions with cryptographic overhead that still remains acceptable. The results of this research lay the groundwork for medical AI systems of the future that are both secure and respectful of patients' privacy.

**Keywords** Smart Healthcare · Internet of Things (IoT) · Fog Computing · Federated Deep Learning (FDL) · Quantum Key Distribution (QKD) · Post Quantum Cryptography (PQC) and · Differential Privacy (DP)

## 1 Introduction

A new and exciting paradigm for distributed machine learning (ML) is federated learning (FL). It shines in fields where patient confidentiality is paramount, including medical [1]. While traditional neural network training methods require consolidation of data at a single training center, Fl can enable multiple separate clients to participate





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49. Ganapathy, G., Anand, S.J., Jayaprakash, M., Lakshmi, S., Priya, B., V., Pandi, V., S: A blockchain based federated deep learning model for secured data transmission in healthcare IoT networks. Measurement: Sens. **33**, 101176 (2024). https://doi.org/10.1016/j.measen.2024.101176
50. Zhao, Z., Li, X., Luan, B., Jiang, W., Gao, W., Neelakandan, S.: Secure Internet of Things (IoT) using a novel Brooks Iyengar quantum Byzantine Agreement-centered blockchain Networking (BIQBA-BCN) model in smart healthcare. Inf. Sci. **629**, 440–455 (2023). https://doi.org/10.1016/j.ins.2023.01.020
51. Liu, Q., Song, Z., Liang, Y., Xie, Z., Zhang, S., Zhang, J., Li, Y.: CoRLHF: Reinforcement learning from human feedback with cooperative policy-reward optimization for LLMs. Expert Syst. Appl. **301**, 130113 (2026). https://doi.org/10.1016/j.eswa.2025.130113
52. Zhang, F., Zhang, C., Guan, J., Zhou, Q., Chen, K., Zhang, X., Du, X.: Breaking the Edge: Enabling Efficient Neural Network Inference on Integrated Edge Devices. IEEE Trans. Cloud Comput. **13**(2), 694–710 (2025). https://doi.org/10.1109/TCC.2025.3559346
53. Jiang, H., Jiang, Z., Tang, W., Xie, Y., Wang, M., Huang, W., Ye, T.: $\mathsf {RobustHealth}$: Non-Interactive Privacy-Preserving System for Heterogeneous Mobile Health Diagnosis. IEEE Trans. Mob. Comput. 1–15 (2025). https://doi.org/10.1109/TMC.2025.3634422
54. Deng, X., Ding, S., Lin, H., Jiang, L., Sun, H., Jin, J.: Privacy-preserving online medical image exchange via hyperchaotic memristive neural networks and DNA encoding. Neurocomputing. **653**, 131132 (2025). https://doi.org/10.1016/j.neucom.2025.131132
55. Long, W., Wang, Y., Long, Q., et al.: Phased-Enhancement Marine Predators Algorithm for Global Optimization and Medical Insurance Fraud Detection. J. Bionic Eng. **23**, 1088–1111 (2026). https://doi.org/10.1007/s42235-026-00845-6

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## Authors and Affiliations

**N. Kannan[1] · K. Balasubramanian[1]**

✉ N. Kannan
n789kannan@gmail.com

K. Balasubramanian
kbala0211@gmail.com

[1] Department of Computer Science and Engineering, E.G.S. Pillay Engineering College, Nagapattinam, India